

SandPro Push Notifications

FIRST POST-MEETING BUILD

Post-meeting implementation plan: ready to build immediately after alignment with Jake and Merci.

Plain-English Promise

We will extend the notification system that already exists: when SandPro creates an in-app alert, the server will also send a focused Web Push alert to opted-in devices, honoring user preferences and opening the right objective when tapped.

BUILD COMPONENTS

What We Would Add

+ Database

push_subscriptions: user, endpoint, keys, device metadata, active/revoked.

+ VAPID Keys

Public key to frontend; private key in Vercel env so the server proves SandPro is sender.

+ Frontend Flow

Settings/onboarding button checks support, asks permission, subscribes, saves to Supabase.

+ Service Worker

Handle push, show notification, handle tap, open objective/deep link; optional badge count.

+ Server Sender

Use web-push, check preferences, send payloads, remove expired device subscriptions.

+ Preference Rules

Push on/off plus channels: mentions, assignments, blockers, at-risk, due/overdue.

SYSTEM FLOW

How It Works

1. App Creates Alert

Mention, assignment, blocker, or due-date event.



2. Preference + Device Check

Confirm push is enabled and find active subscriptions.



3. Web Push Delivery

Backend sends title, body, and secure URL payload.



4. User Taps Notification

Phone/browser opens SandPro to the right objective.

Technical basis: service worker + PushManager.subscribe() + server-side Web Push.

ROLLOUT

Best Implementation Plan

Phase 1

Subscriptions, service worker push handler, Enable Push button, mentions + assignments.

Phase 2

Blockers, at-risk, due, and overdue push with clear preference controls.

Phase 3

Mobile install guidance and QA on iPhone and Android devices.

Phase 4

Tune noise rules with Jake/Merci after real usage.

Mobile Reality

- **Android/Chrome:** straightforward after the user allows notifications.
- **iPhone:** Web Push works for Home Screen web apps on iOS/iPadOS 16.4+; user must add SandPro to Home Screen and grant permission from a tap.
- **Limit:** pop-up/disappear behavior is partly OS-controlled; we can reduce noise but not guarantee exact persistence behavior.

Meeting takeaway: This is not a tiny CSS change, but it fits cleanly into SandPro's existing in-app/email notification foundation and is ready to start as the first post-meeting build.